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SIRUSERI - CHENNAI



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PROJECT X 2026



- Problem Statement ID – XYZON-01
- Problem Statement Title- Decentralized Credential Verification System
- PS Category- Software/Hardware ✓
- Team Name – Wolf
- Team Leader Name – Mohamed Wasim





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PROBLEM STATEMENT



In India's vocational education system, certificate verification is often manual, time-consuming and prone to forgery. This reduces trust among employers and creates challenges in validating candidates' skills



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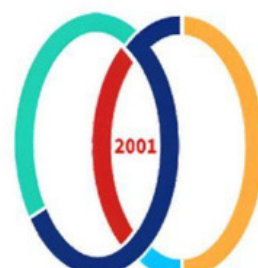
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PROPOSED SOLUTION



We propose CrediChain, a blockchain-inspired decentralized credential verification system that ensures certificates are tamper-proof, instantly verifiable, and secure.

The system eliminates manual verification and prevents forgery by using cryptographic hashing and immutable ledger storage.

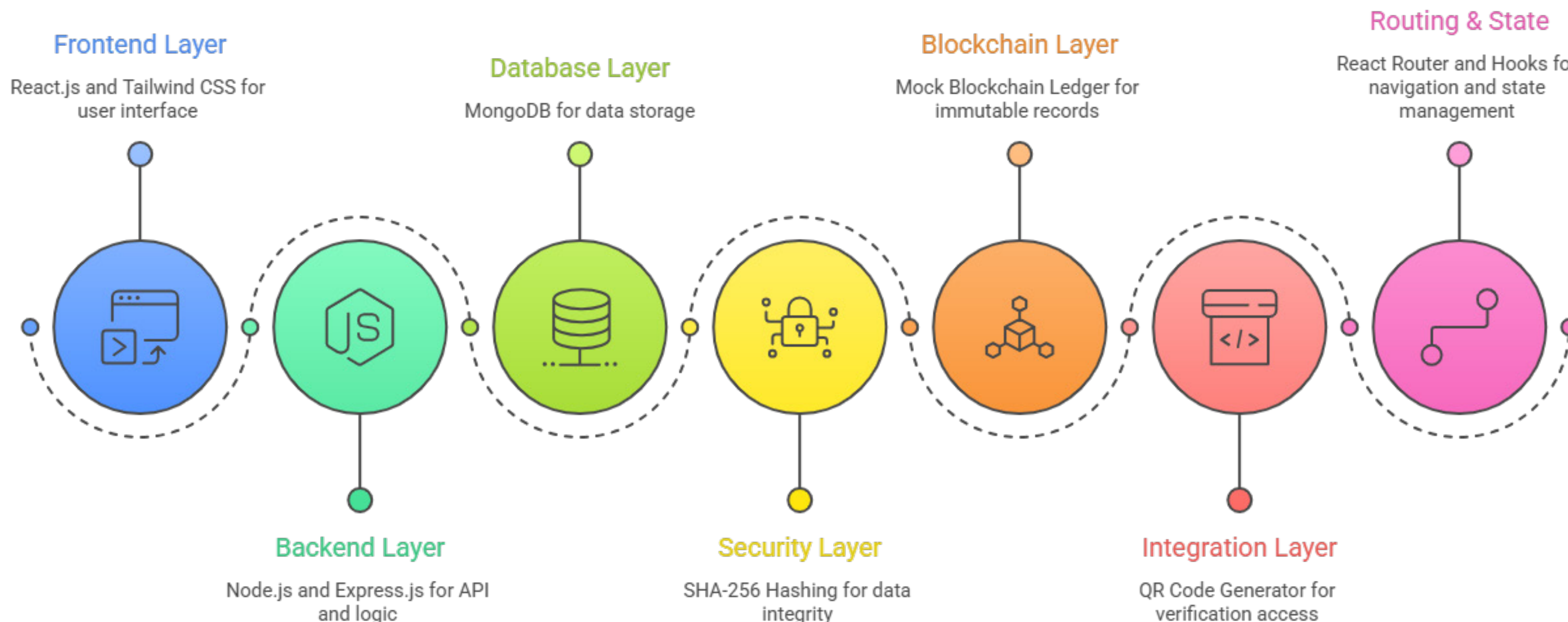


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TECH STACK



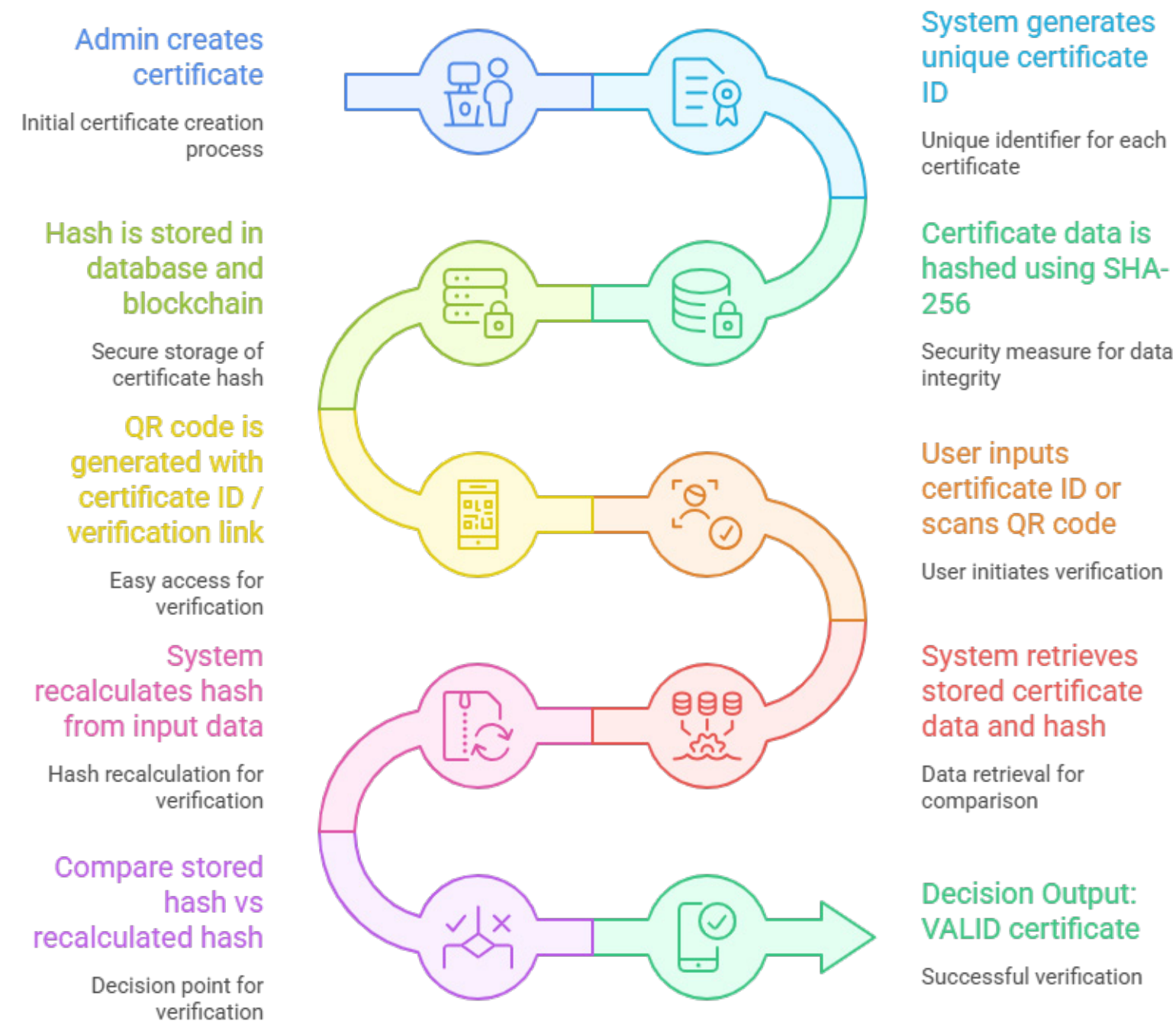
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ARCHITECTURE / WORKFLOW



CrediChain Certificate Verification Workflow



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FEATURES AND INNOVATION



Features:

- Secure certificate generation with unique ID
- SHA-256 hashing for data integrity
- Blockchain-inspired immutable storage
- QR code-based instant verification
- Verification via ID or QR (Valid / Tampered / Not Found)
- Real-time dashboard with verification stats

Innovations:

- Real-time tamper detection using hash comparison
- Detects even smallest data modification
- Lightweight blockchain approach (fast & efficient)
- QR-driven seamless verification experience
- Role-based system (Admin & Verifier)
- Fraud/tamper attempt tracking



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FEASIBILITY AND VIABILITY



Feasibility:

- Uses simple, widely available tech stack
- Easy implementation with SHA-256 hashing
- Mock blockchain reduces complexity
- Low cost and fast development

Viability:

- Solves real issue of certificate forgery
- Useful for institutions, employers, and recruiters
- Reduces manual verification time
- Scalable for real-world deployment



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RESEARCH AND REFERENCES

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PROJECT
2026

- Study of cryptographic hashing (SHA-256) for data integrity
- Basics of blockchain and immutable ledgers
- Research on digital certificate verification systems
- Documentation of Node.js, React, MongoDB
- Libraries:
 - crypto (Node.js)
 - qrcode (npm)
- Online resources:
 - MDN Web Docs
 - GeeksforGeeks



IMPLEMENTATION / DEMO



ENTERPRISE CONSOLE

Manage certificate issuance, verification, and audit trails.

Verify Certificate

Scan or enter ID

GO

VISUAL LANGUAGE

Primary

CrediChain

CERTIFICATE VERIFICATION SYSTEM

VERIFICATION CONSOLE

Verify any certificate in one click.

Paste a certificate ID or scan the QR code. The system fetches the stored record, recalculates the SHA-256 hash, and compares it against the blockchain record.

Paste certificate ID or QR verification URL

Verify Certificate

LOAD CRD-DEMO-001

LOAD CRD-DEMO-002

SIMULATE QR SCAN

ROLE: VERIFIER

Logout

UPLOAD AND VERIFY EXISTING CERTIFICATE

JSON upload or manual input

Submit a certificate payload. CrediChain regenerates the hash and compares it with stored records.

CERTIFICATE JSON (OPTIONAL)

{"certificateId":"CRD-DEMO-001","name":"Ava Johnson","course":"Advanced Blockchain Development","issueDate":"2026-04-14"}

CERTIFICATE ID

NAME

COURSE

ISSUE DATE

15-04-2026

Upload and Verify

UPLOAD VERIFICATION RESULT

No upload verification yet.

ENTERPRISE CONSOLE

Manage certificate issuance, verification, and audit trails.

Dashboard

Overview & KPIs

GO

Issue Certificate

Create and sign

GO

CrediChain

CERTIFICATE VERIFICATION SYSTEM

CREDICHAIN - DECENTRALIZED CREDENTIAL VERIFICATION SYSTEM

Tamper-proof certificates. Instant trust.

A blockchain-inspired credential network where every certificate can be validated in seconds using SHA-256 hash comparison and immutable ledger records.

AUTHENTICITY

INSTANT VERIFICATION

ZERO FORGERY

ROLE: ADMIN

Logout

VISUAL LANGUAGE

Primary

Success

Error

TOTAL CERTIFICATES

12

BLOCKCHAIN RECORDS

12

VERIFICATIONS

17

VALID CHECKS

16

TAMPERED ATTEMPTS

0

FRAUD ATTEMPTS

1

FRAUD ATTEMPTS

1

Invalid verification attempts detected

VERIFICATION SUCCESS RATE

94%

Valid checks over total checks

AVERAGE VERIFICATION TIME

< 1s

Fast hash compare with ledger lookup

PROBLEM

Certificate forgery is common

Fake credentials are easy to produce and hard to detect quickly in manual processes.

PROBLEM

Manual verification is slow

Recruiters and institutions spend too much time validating records over email and paperwork.

PROBLEM

No trusted digital trail

Most certificates lack a tamper-proof audit path that can be verified instantly.

@Project X idea submission- Template



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1. Kindly keep the maximum slides limit up to Eight **(8)**. (Including the title slide)
2. Try to avoid paragraphs and post your idea in points /diagrams / Infographics /pictures
3. Keep your explanation precise and easy to understand
4. Idea should be unique and novel.
5. You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).
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